Punam Shrestha: shresthapunam@gmail.com

Padel Game Analytics

Shot Classification System Using Computer Vision & Machine Learn

Abstract

This project presents the design and development of an automated shot classification system for padel gameplay analysis using Computer Vision and Machine Learning techniques. The system processes recorded match videos to detect and track players, rackets, and the ball, subsequently classifying each shot as a forehand, backhand, or smash/serve. Leveraging the YOLO object detection framework, OpenCV for video processing, and optional MediaPipe pose estimation, the system generates structured analytical output in CSV and JSON formats, enabling coaches and players to review performance data efficiently. The prototype demonstrates the practical feasibility of applying AI-driven sports analytics to a fast-paced racket sport and establishes a foundation for more advanced real-time analysis capabilities.

1. Introduction

The rapid advancement of Artificial Intelligence (AI) and Computer Vision over the past decade has opened new frontiers in sports performance analysis.Automated video analysis systems now offer the ability to process large volumes of match footage objectively, extracting quantitative insights that were previously inaccessible.

Padel is one of the fastest-growing racket sports globally, combining elements of tennis and squash within an enclosed court. Its dynamic nature, characterised by rapid exchanges, varied shot techniques, and strategic wall play, makes manual performance evaluation particularly challenging. Coaches and players require detailed, repeatable, and objective data on shot selection and technique in order to improve competitive performance.

This project addresses that requirement by developing a Computer Vision and Machine Learning pipeline capable of ingesting padel gameplay video footage and producing structured, shot-level analytics. The system detects key game objects — players, rackets, and the ball — tracks their trajectories across frames and classifies each recorded shot according to its type. The resulting output is exported in both CSV and JSON formats, making the data readily consumable by downstream analysis tools.

The project demonstrates how well-established technologies, including the YOLO object detection framework, OpenCV, and optional Media Pipe-based pose estimation, can be integrated into a cohesive sports analytics prototype. Beyond its immediate application to padel, the methodology established herein is transferable to a range of racket and court sports, offering broad potential for future development.

2. Problem Statement

Padel is a fast-paced racket sport in which multiple shot types are executed at high speed, often in rapid succession. Manual analysis of match footage to identify and categories individual shots is an extremely labor-intensive task, demanding considerable time and domain expertise from coaching staff. Furthermore, human observation introduces subjectivity and inconsistency, limiting the reliability of performance assessments conducted across multiple sessions or between different analysts.

Existing commercial sports analysis platforms are predominantly designed for mainstream sports such as football, basketball, and tennis, leaving padel largely underserved in terms of automated analytical tooling. Coaches and players therefore lack access to affordable, purpose-built solutions that can automatically process video footage and deliver structured, shot-level performance data.

This project aims to address this gap by developing a prototype automated system that processes padel match video and produces objective, structured analytics. Specifically, the system seeks to detect the primary game objects within each video frame, track the continuous motion of players and the ball, recognise distinct shot types based on motion characteristics and player pose, and output the classified data in standard machine-readable formats. In doing so, the project provides a practical foundation upon which a full-featured, production-quality padel analytics platform could be constructed.

3. Objectives

The primary objective of this project is to design and implement a functional prototype system for automated padel shot classification from gameplay video. In pursuit of this overarching goal, the following specific objectives have been defined:

- To develop a video ingestion and frame extraction pipeline capable of processing standard padel match recorded at typical broadcast or smartphone resolutions.
- To implement an object detection module using the YOLO framework to reliably identify and localize players, rackets, and the ball within individual video frames.
- To design and integrate an object tracking mechanism that maintains consistent identities for detected objects across consecutive frames, thereby enabling trajectory and motion analysis.
- To develop a shot classification algorithm that leverages ball trajectory, racket motion, and player body pose to categorise each detected shot as a forehand, backhand, or smash/serve.

- To generate structured output files in both CSV and JSON formats containing per-shot analytics, including frame number, timestamp, shot type, and associated player identifier.
- To evaluate the performance of the implemented system against manually labelled test footage and identify key areas for future improvement.

4. Methodology

The development of the system followed a structured, pipeline-based methodology in which each processing stage builds incrementally upon the outputs of the preceding stage. This approach ensures modularity, facilitates independent testing of individual components, and simplifies future extension or replacement of any given stage.

4.1 Video Input and Frame Extraction

The system accepts a pre-recorded padel match video as its primary input. Video capture and frame-by-frame extraction are handled using the OpenCV library, which provides a robust and widely supported interface for reading video files encoded in standard formats such as MP4 and AVI. Each extracted frame is passed individually to the subsequent object detection stage, ensuring that the system operates at the granularity of a single frame and thereby captures transient events such as ball contact.

4.2 Object Detection

Object detection is performed using the YOLO (You Only Look Once) architecture, accessed through the Ultralytics API. YOLO is a single-stage detector that processes each frame in its entirety and predicts bounding boxes and class labels for all detected objects in a single forward pass, making it well-suited to real-time or near-real-time applications. The model is configured to detect three object classes relevant to padel analysis: players, rackets, and the ball. Each detected object is characterized by its bounding box coordinates, confidence score, and class label.

4.3 Object Tracking

Following detection, a tracking module is employed to associate detected objects across consecutive frames and maintain consistent identities throughout the duration of the video. Tracking is achieved by analysing spatial continuity between bounding boxes in successive frames, using motion prediction to handle brief occlusions or detection failures. This stage produces a set of trajectories for each tracked object, which serves as the primary input for the subsequent shot classification module. Accurate tracking is particularly critical for the ball, given its small size, high velocity, and frequent periods of occlusion.

4.4 Pose Estimation (Optional)

To supplement ball and racket trajectory data, the system optionally integrates MediaPipe Pose, a lightweight pose estimation framework capable of detecting key body landmarks in real time. Player pose information — specifically the positions of the shoulder, elbow, wrist, and hip landmarks — provides valuable contextual cues for distinguishing between shot types that may exhibit similar ball trajectories, such as forehand and backhand shots. This module is implemented as an optional component to preserve processing efficiency when pose data is not required.

4.5 Shot Classification

Shot classification is performed by analysing the combined features derived from ball trajectory, racket motion vector, and optional player pose data at the moment of contact. Each detected contact event is assigned one of three categories based on predefined heuristic rules informed by the biomechanical characteristics of padel shot types. A forehand is identified by a swing originating from the player's dominant side, a backhand by a cross-body swing motion, and a smash or serve by a rapid overhead racket motion accompanied by a steeply descending ball trajectory. The classification logic is designed to be extensible, allowing additional shot categories to be incorporated in future iterations.

4.6 Data Storage and Output

The results of the classification stage are aggregated and serialised into two structured output formats. A CSV file provides a tabular record suitable for import into spreadsheet applications and statistical analysis tools, while a JSON file provides a hierarchical, machine-readable format appropriate for integration with web applications or databases. Each output record contains the frame number, the corresponding video timestamp, the classified shot type, and the identifier of the player who executed the shot.

5. System Design and Architecture

The system is designed as a sequential, modular processing pipeline in which each component performs a discrete transformation on the data flowing through it. This architecture promotes separation of concerns, enabling individual components to be developed, tested, and replaced independently without disrupting the overall workflow. The pipeline comprises six principal stages, as described below.

Stage	Component	Description
1	Video Input	Reads padel match footage frame by frame using OpenCV VideoCapture.
2	Frame Extraction	Decodes individual frames and prepares them for object detection.
3	Object Detection	Runs YOLO inference to detect players, rackets, and the ball per frame.
4	Object Tracking	Associates detections across frames to form continuous trajectories.
5	Pose & Motion Analysis	Analyses player pose (MediaPipe) and racket/ball motion vectors.
6	Shot Classification	Classifies each contact event into forehand, backhand, or smash/serve.
7	Output Generation	Serialises classified shot data to CSV and JSON output files.

The modular design ensures that any stage can be upgraded — for example, replacing the heuristic shot classifier with a trained

Punam Shrestha: shresthapunam@gmail.com

deep learning model — without modifying the surrounding pipeline. All inter-stage communication is performed via in-memory data structures, minimising I/O overhead during processing.

6. Implementation

The system is implemented entirely in Python, leveraging its extensive ecosystem of scientific computing and computer vision libraries. The implementation adheres to a class-based, object-oriented design where appropriate, and uses a procedural pipeline structure for the main processing loop.

6.1 Video Capture

Video input is managed using the cv2.VideoCapture interface provided by OpenCV. The capture object reads frames sequentially from the specified input file path. Frame-level metadata, including the frame index and elapsed time calculated from the video's frames-per-second (FPS) parameter, is extracted at this stage and associated with all subsequent analytical outputs.

6.2 YOLO Object Detection

The Ultralytics YOLO library is used to load a pretrained detection model and perform inference on each extracted frame. The model produces a list of detected instances per frame, each described by a bounding box in pixel coordinates, a class identifier, and a detection confidence score. Detections below a configurable confidence threshold are discarded to reduce false positive noise.

6.3 Multi-Object Tracking

A lightweight tracking mechanism maintains object identities across frames by associating each new detection with the closest matching detection from the preceding frame, subject to a maximum displacement threshold. Players are assigned persistent identifiers (Player 1, Player 2) based on their initial detection positions, and these identifiers are propagated throughout the tracking lifecycle.

6.4 Shot Classification Logic

The shot classification module evaluates a temporal window of tracking data centred on each detected ball-racket contact event. The relative horizontal displacement of the racket bounding box between successive frames determines the swing direction, while the vertical velocity of the ball at the moment of contact is used to distinguish overhead shots from ground strokes. Where MediaPipe pose data is available, the wrist-to-shoulder angle is additionally considered to improve classification accuracy for edge cases.

6.5 Output Generation

Classified shot records are accumulated in a list of dictionaries throughout processing. Upon completion of the video, the Pandas library is used to export the accumulated data as a CSV file. The same data is simultaneously serialised to JSON using Python's built-in json module, ensuring that both output formats remain synchronised. Output files are written to a user-configurable output directory.

7. Tools and Technologies Used

The following tools, frameworks, and libraries constitute the technical stack of the implemented system. Each was selected on the basis of its suitability to the specific task it performs, its maturity and community support, and its compatibility with the overall Python-based architecture.

Technology / Library	Version	Purpose
Python	3.10+	Primary programming language for all system components.
OpenCV (cv2)	4.x	Video capture, frame extraction, and image preprocessing.
Ultralytics YOLO	8.x	Real-time object detection for players, rackets, and the ball.
MediaPipe	0.10+	Optional skeletal pose estimation for player body landmark detection.
NumPy	1.24+	Numerical array operations and coordinate geometry calculations.
Pandas	2.x	Structured data management and CSV file export.
Matplotlib	3.x	Visualization of trajectories, bounding boxes, and analytics charts.
PyTorch / TensorFlow	Latest	Deep learning backend for YOLO model inference.
JSON (built-in)	—	Serialization of shot classification results to JSON format.

8. Results and Output

Upon processing a padel match video, the system produces structured analytical output capturing each classified shot event. The results demonstrate the system's ability to identify distinct shot types and associate them with the corresponding player and temporal position within the footage. The following subsections present representative sample output in both supported formats.

8.1 CSV Output

The CSV output provides a tabular summary of all detected shot events, with one row per event. The table below presents a representative excerpt of this output.

Frame	Timestamp	Shot Type
120	00:00:04	Forehand
250	00:00:08	Smash
378	00:00:12	Backhand
490	00:00:16	Forehand
610	00:00:20	Backhand

8.2 JSON Output

The JSON output mirrors the information contained in the CSV file but in a hierarchical, nested format suitable for consumption by web services, databases, or further programmatic processing. Each shot event is represented as a self-contained JSON object

Punam Shrestha: shresthapunam@gmail.com

within an array, as illustrated in the following representative excerpt.

[{ "frame": 120, "timestamp": "00:00:04", "shot_type": "Forehand", },

{ "frame": 250, "timestamp": "00:00:08", "shot_type": "Smash"}]

8.3 Observed Performance

During initial testing on sample footage, the system successfully detected and tracked both players and the ball across the majority of frames under standard lighting conditions and fixed camera angles. Shot classification yielded satisfactory accuracy for forehand and smash events, where distinctive racket motion patterns provided strong discriminating features. Backhand classification exhibited slightly lower precision due to the visual similarity between certain backhand and forehand swing trajectories when viewed from non-optimal camera angles. Overall, the prototype demonstrated sufficient functionality to validate the proposed architecture and establish a meaningful baseline for future improvement.

9. Challenges and Limitations

The development and testing of the system revealed several technical challenges and limitations inherent to the problem domain. These are documented below to inform future development efforts and provide an honest assessment of the prototype's current capabilities.

- Fast Ball Movement: The padel ball travels at high speed during smashes and serves, frequently resulting in motion blur that degrades detection accuracy. Standard YOLO models trained on static imagery may struggle to localise a severely blurred ball, leading to missed detections during high-velocity exchanges.
- Small Object Detection: The padel ball occupies a very small proportion of the total frame area, particularly in wide-angle or full-court views. Small object detection remains a known limitation of general-purpose detectors, and performance on the ball was noticeably lower than on the larger player and racket objects.
- Occlusion: Players frequently obstruct the camera's view of the ball and each other, particularly during net exchanges and volleys. Occlusion causes tracking interruptions that can result in identity switches or lost trajectories, reducing the completeness of the output analytics.
- Camera Angle Dependency: The accuracy of shot classification is sensitive to the camera angle from which the footage was recorded. Side-on views provide the clearest differentiation between forehand and backhand swings, whereas end-on or elevated angles reduce the visibility of horizontal racket motion.
- Heuristic Classification Limitations: The current shot classification module relies on hand-crafted heuristic rules rather than a trained machine learning classifier. While this approach is interpretable and requires no labelled training data, it is inherently less robust and generalisable than a data-driven model.
- Computational Cost: Processing full-resolution video through YOLO

Punam Shrestha: shresthapunam@gmail.com

inference on a CPU is significantly slower than real time. GPU acceleration substantially improves throughput but introduces hardware requirements that may not be universally available.

10. Future Improvements

The current prototype establishes a viable foundation for a fully featured padel analytics platform. A number of enhancements are proposed to address the identified limitations and extend the system's capabilities in future iterations.

- Real-Time Processing: Optimising the pipeline for real-time operation — through hardware acceleration, model quantisation, and inference batching — would enable live match analysis and immediate post-point feedback during training sessions.
- Deep Learning-Based Shot Classification: Replacing the heuristic classifier with a temporal action recognition model trained on labelled padel video data would significantly improve classification accuracy, particularly for ambiguous and edge-case shots.
- Expanded Shot Categories: The current system recognises three shot types. Future versions could classify a broader vocabulary of padel shots, including the bandeja, vibora, lob, and drop shot, providing more granular tactical insights.
- Player Performance Statistics: Aggregating shot-level data over the course of a full match would enable the generation of player-level statistics, such as shot frequency distributions, rally length, and winning shot analysis, supporting in-depth performance review.
- Heatmap Visualization: Overlaying player positioning and shot-execution locations onto a court diagram as a heatmap would provide an intuitive visual representation of court coverage and tactical patterns.
- Multi-Camera Support: Integrating footage from multiple synchronized camera angles would eliminate occlusion-related tracking failures and provide richer multi-view features for classification, at the cost of increased system complexity.
- Custom Training Data: Fine-tuning the YOLO model on a domain-specific padel dataset would improve detection accuracy for both the ball and racket, addressing the small object detection challenge identified during testing.

11. Conclusion

This project has successfully demonstrated the application of Computer Vision and Machine Learning techniques to the automated analysis of padel gameplay footage. By integrating the YOLO object detection framework with OpenCV-based video processing, a custom tracking mechanism, and a heuristic shot classification module, the system is capable of ingesting match video, detecting and tracking the principal game objects, classifying shot events by type, and producing structured analytical output in CSV and JSON formats.

The prototype validates the technical feasibility of the proposed approach and provides a clear, modular architecture upon which a more sophisticated analytics platform can be constructed. The challenges encountered during development — including fast ball motion, small object detection, and occlusion — are well-understood problems within the Computer Vision literature, and credible solutions exist for each. The path from the current prototype to a production-quality system is therefore well-defined.

In a broader context, this project illustrates how modern AI tooling has substantially lowered the barrier to entry for sports analytics, enabling domain-specific analysis systems to be developed within the scope of a student project with meaningful and demonstrable outcomes. As padel continues its global expansion, automated analytical tools of this nature are likely to become increasingly valuable to coaches, players, and analysts seeking a competitive edge through data-driven performance improvement.

12. References

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